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Sinusitis

Definition

- ☑ Acute sinusitis is an illness that results from infection of one or more of the paranasal sinuses.
- ☑ By the duration of clinical symptoms, sinusitis is called acute, when symptoms completely resolve in <30 days.

Difference Between Acute Bacterial Sinusitis and Acute Viral Sinusitis

Clinical characteristics	***Acute bacterial sinusitis***	***Acute viral sinusitis***
Fever	Fever may develop or recur on day 6 to 7 of illness after initial improvement, high grade since start	Fever is present in earlier part of illness (initial 1–2 days), mild grade
Nasal discharge	Fails to improve substantially or worsens over time	Peaks on Days 3 to 6, then steadily improves
Cough	Fails to improve substantially or worsens over time	Peaks on Days 3 to 6, then steadily improves
Ill appearance	Present with severe disease	Absent
Headache	Present with severe disease	Absent
Clinical course	Symptoms are present for ≥10 days without improvement	Symptoms peak in severity on Days 3 to 6 and then improve

Microbiology

- ☑ *H. influenzae* (non-typeable), *Streptococcus pneumoniae*, and *Moraxella catarrhalis* are the major pathogens in uncomplicated ABS in otherwise healthy children.
- ☑ In complicated ABS polymicrobial infections are common with dominance of *Streptococcus* species (e.g., *Streptococcus* anginosus group), other anaerobes, and *Staphylococcus* species.

Clinical Manifestations

ABS has three characteristic presentations.

Clinical presentation	***Description***
Persistent symptoms	Nasal discharge or cough or both for >10 days without improvement
Severe symptoms	Onset with temperature of ≥39°C (102.2°F) and purulent nasal discharge for ≥3 consecutive days
Worsening symptoms	Respiratory symptoms (nasal discharge or cough, or both) that worsen after initial improvement, or onset of new fever or severe headache

Diagnosis

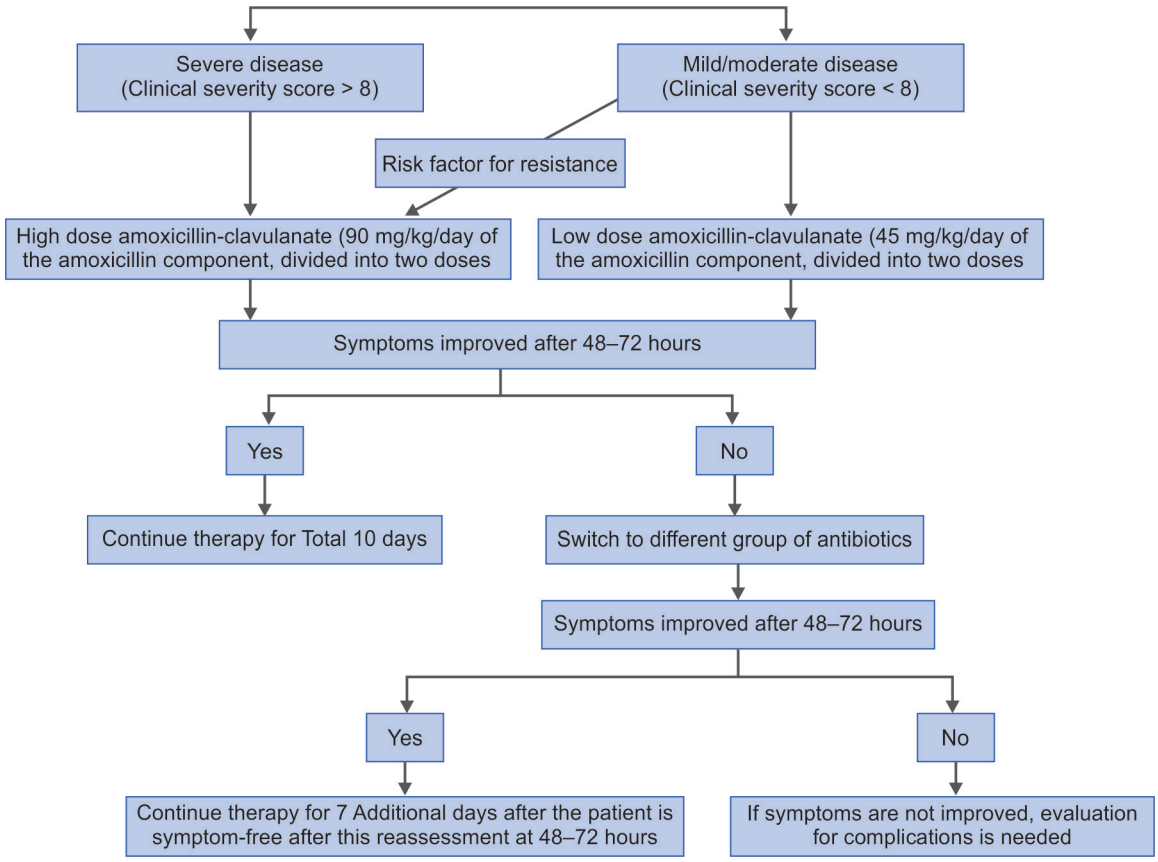
- ☑ History and based on stringent clinical criteria, including persistent clinical features of an upper respiratory infection beyond 10 days, without much improvement; an upper respiratory infection with high fever and purulent nasal discharge at onset lasting for at least three consecutive days; and biphasic or worsening symptoms (double sickening).

Clinical Severity Score for Acute Bacterial Sinusitis

A total score <8 indicates mild/moderate disease. A total score ≥8 indicates severe disease.

Symptom or sign	***Points***
Abnormal nasal or postnasal discharge:	
Minimal	1
Severe	2
Nasal congestion	1
Cough	2
Malodorous breath	1
Facial tenderness	3
Erythematous nasal mucosa	1
Fever:	
<38.5°C	1
≥38.5°C	2
Headache (retro-orbital)/irritability:	
Severe	3
Mild	1

Treatment



Chronic Sinusitis in Children

Chronic sinusitis is characterized by persistent inflammation of the mucosa of paranasal sinuses. Pediatric chronic rhinosinusitis (CRS) is more prevalent than acute rhinosinusitis and similar to allergic rhinitis. Prevalence is most common among 2–6 year and gradually decline thereafter.

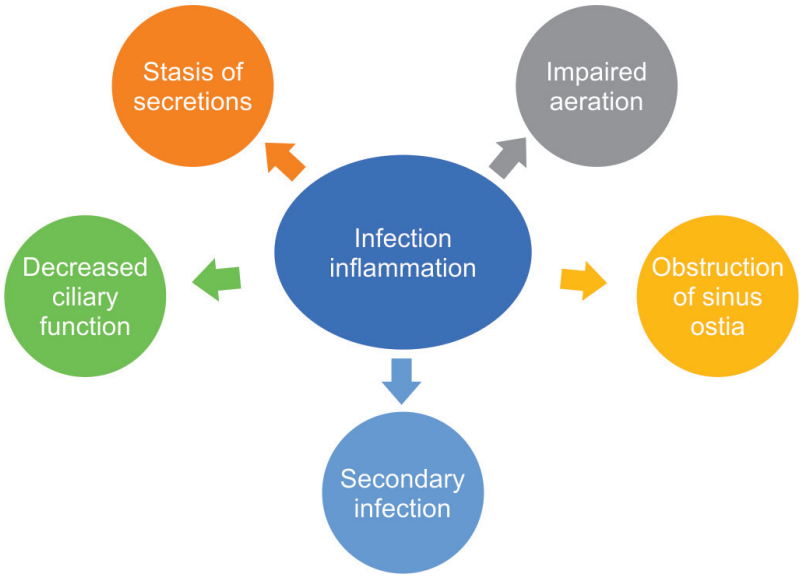
Definition

American Academy of Otolaryngology—Head and Neck Surgery defined pediatric CRS as at least 90 continuous days of 2 or more symptoms of purulent rhinorrhea, nasal obstruction, facial pressure/pain, or cough and either endoscopic signs of mucosal edema, purulent drainage, or nasal polyposis, and/or CT scan showing mucosal changes within the ostiomeatal complex and/or sinuses.

It is classified into 2 main groups of **with or without nasal polyposis**. Patients with nasal polyposis have different pathophysiology and management, and have increased risk of underlying conditions such as cystic fibrosis, allergic fungal sinusitis and antrochoanal polyps.

Etiopathogenesis remains unclear and may be an inappropriate or excessive immune response to an external stimulus inhaled through nasal airway. Viral or bacterial infection along with comorbid conditions and environmental triggers leads a cascade of process leading to chronic sinusitis.

Conditions Associated with Chronic Sinusitis



Anatomical defects	*Chronic inflammation*
☑ *Intrinsic defects:* Septal deviation, nasal polyps, lateral wall anomalies such as concha bullosa, paradoxical middle turbinates, etc. ☑ Contiguous spread from dental region ☑ *Extrinsic defects:* Foreign bodies such as nasogastric and nasotracheal tubes	☑ Allergic rhinitis* ☑ Allergic fungal sinusitis ☑ Gastroesophageal reflux,* cystic fibrosis, primary immunodeficiencies, primary ciliary dyskinesia, asthma, environmental factors such as tobacco smoke, air pollutants

*controversial role

Common microbiological pathogens: Generally polymicrobial infections are suggested.
Alpha-hemolytic *Strep, Staph. aureus,* coagulase negative *Staph. aureus, Streptococcus pneumonia*, non-typeable *H. influenza, Moraxella catarrhalis,* anerobes
P. aeruginosa and other aerobic gram-negative rods: More common nosocomial origin or immuno-compromised or those with CF
Fungi pathogens: Bipolaris, Curvularia, Aspergillus, and *Dreschlera* species. Children with nasal polyposis are predisposed to allergic fungal sinusitis (AFS) whereas immunocopromised children may have invasive fungal disease

Evaluation of Children with Chronic Sinusitis

History

Nonspecific and difficult to distinguish from recurrent viral upper respiratory infections, chronic allergic rhinitis, or chronic nasal congestion due to adenoid hypertrophy.

Symptoms

Chronic nasal congestion, purulent rhinorrhea, facial pain and pressure, and cough. Mouth breathing and otitis media are also common. Hyposmia and headache are less common.

Complications

Bony erosion and expansion due to polyps or mucoceles, sclerosis due to chronic osteitis, and recurrent pulmonary exacerbations in the setting of asthma or cystic fibrosis

Evaluation of Children with Chronic Sinusitis

Investigations

- ☑ *Plain radiographic studies:* Not recommended for either diagnosis or follow-up.
- ☑ *Computed tomography (CT):* Imaging of choice though not routinely recommended.
- ☑ *Contrast CT scan/MRI:* Reserved for suspected abscess formation in orbit/ brain.
- ☑ *Nasal endoscopy:* Flexible or rigid for visualization of purulent discharge, adenoid hyperplasia or infection, nasal polyps, mucosal edema, and septal deviation.
- ☑ *Cultures:* Only if no response to empiric therapy within 72 hours, significant comorbidities, or severe illness. *Maxillary sinus aspirations or endoscopically-guided middle meatal cultures are preferred over blind paranasal swabs.*

Medical Treatment

Aim: Eradicate bacterial infection and reduce underlying sinonasal inflammation.

Mainstay:

- ☑ Oral antibiotics (Amoxicillin, amoxicillin/clavulanate (45 mg/kg per day divided every 12 hours), or cephalosporin are 1st line.; Clindamycin (20–40 mg/kg per day orally divided every 6–8 hours) for anaerobes and in penicillin-allergic, dual therapy with 3rd generation cephalosporin and clindamycin or levofloxacin is recommended. *Duration of therapy:* No consensus, upto 10–20 days)
- ☑ Daily nasal saline irrigation/spray and nasal steroids along with antibiotics are 1st line treatment.

Adjunctive Treatment

- ☑ Oral steroids (*short course:* 2 weeks 1–2 mg/kg prednisolone or equivalent) in children with severe symptoms, polyps, allergic fungal sinusitis)
- ☑ *Antifungals:* Allergic fungal disease. Most preferred Itraconazole (5–10mg/kg)
- ☑ Antihistamines (second generation), and anti-leukotrienes (aspirin sensitivity/ allergic/asthma with nasal polyps)
- ☑ *In allergic rhinitis:* Allergen avoidance, anti-histamines (second generation), nasal steroids and allergen immunotherapy.
- ☑ *Controversial:* Role of empiric anti-reflux medication, topical antibiotics, and antral irrigation

ENT surgeon should be consulted in patients with suspected underlying anatomic abnormalities polyps, recurrent disease, failure of medical treatment

Surgical Treatment

In cases failure of maximal medical therapy, comorbidities like underlying anatomical defects, polyps, orbital or intracranial complications, underlying diseases such as immunodeficiencies, cystic fibrosis, ciliary dyskinesia, uncontrolled asthma, etc.

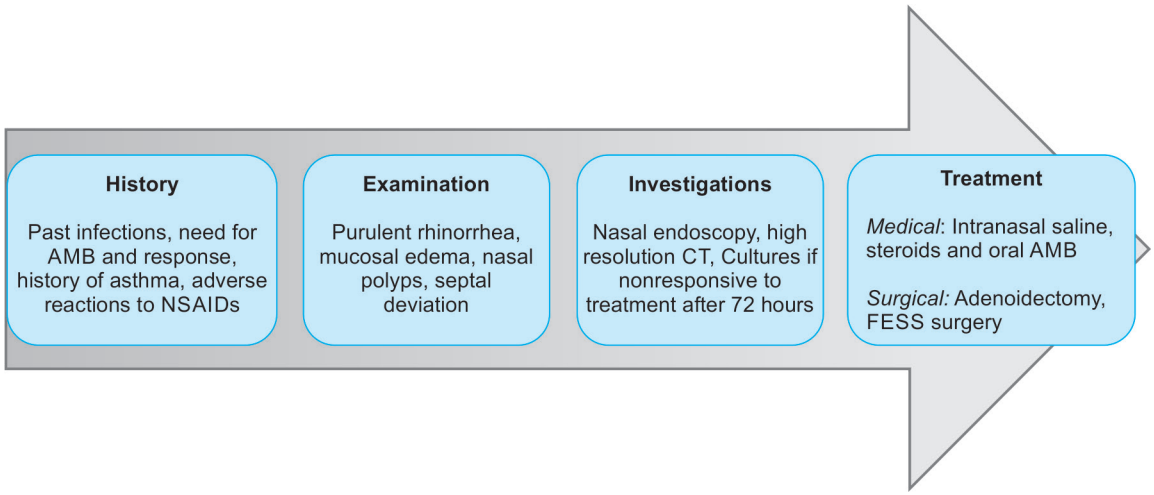
Aim: Eradicate potential bacterial reservoirs and enhance sinonasal aeration and drainage.

Secondary: Reserved for those failing maximum medical management.

Most effective: Adenoidectomy.

Other techniques: Endoscopic sinus surgery, sinus puncture and lavage, balloon sinuplasty, open surgical approaches and turbinate reduction.

Stepwise Approach to Pediatric Chronic Sinusitis



Further Reading

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Acute Pharyngitis/Acute Tonsillopharyngitis

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